

Project Code: ParentalLoggerViewer- Form1.vb

```

Imports System.IO.packaging
Imports System.io
Public Class Form1
    Private imageCount As Integer = -1
    Private imageNum, clearLocation As Integer
    Private loc As Integer = -1
    Private dirInfo() As String
    Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MyBase.Load

        End Sub

    Private Sub Button3_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles Button3.Click
        'Button3 is text file Browse Button
        OpenFileDialog1.ShowDialog()
        TextBox1.Text = OpenFileDialog1.FileName.ToString
        If (TextBox1.ToString.Contains(".txt") Or
TextBox1.ToString.Contains(".log")) Then
            'load txt file into richtextbox
            Dim line As String
            FileOpen(1, TextBox1.Text, OpenMode.Input, OpenAccess.Read)
            While Not EOF(1)
                line = LineInput(1)
                RichTextBox1.Text += line + Chr(13) + Chr(10)
            End While
            FileClose(1)
        End If

        If TextBox2.Text = "" Or PictureBox1.Image Is Nothing Then
            MsgBox("Please select an image folder by clicking the right
Browse button", MsgBoxStyle.OkOnly)
        Else
            highlightText()
        End If
    End Sub

    Private Sub Button4_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles Button4.Click
        'Button4 is image folder browse button
        FolderBrowserDialog1.ShowDialog()
        TextBox2.Text = FolderBrowserDialog1.SelectedPath()
        If (TextBox2.Text.Contains(".zip") Or TextBox2.Text = "") Then
            MsgBox("Please unzip and select the unzipped logScreenCaptures
folder", vbOKOnly, "Folder Error")
        Else
            'load image into PictureBox1
            dirInfo = System.IO.Directory.GetFiles(TextBox2.Text)

            Dim img As Image
            If dirInfo(0).EndsWith(".jpg") Then
                img = Image.FromFile(dirInfo(0))
            End If
        End If
    End Sub
End Class

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        imageCount = 0
        Dim bmp As New Bitmap(img, PictureBox1.Width,
PictureBox1.Height)
        PictureBox1.Image = bmp
    Else
        MsgBox("Please select a different folder.", vbOKOnly, "Folder
Error")
    End If

End If

If TextBox1.Text <> "" Or RichTextBox1.Text <> "" Then
    highlightText()
Else
    MsgBox("Please select a text file by clicking the left Browse
button", MsgBoxStyle.OkOnly)
End If

End Sub

```

```

Private Sub Button2_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles Button2.Click
    'Button2 is the "next" arrow button
    If PictureBox1.Image Is Nothing Then
        MsgBox("Please select an image folder by clicking the right
Browse button", MsgBoxStyle.OkOnly)
    Else
        'get next image
        If imageCount < UBound(dirInfo) And imageCount > -1 Then
            Dim img As Image
            imageCount += 1
            img = Image.FromFile(dirInfo(imageCount))
            Dim bmp As New Bitmap(img, PictureBox1.Width,
PictureBox1.Height)
            'set PictureBox1 to the next image
            PictureBox1.Image = bmp
        End If
        If TextBox1.Text <> "" Or RichTextBox1.Text <> "" Then
            'highlight word in richtextbox1 that is related to the image
            highlightText()
        End If
    End If
End Sub

```

```

Private Sub Button1_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles Button1.Click
    'Button1 is the "previous" arrow button
    If PictureBox1.Image Is Nothing Then
        MsgBox("Please select an image folder by clicking the right
Browse button", MsgBoxStyle.OkOnly)
    Else
        'get previous image
        If imageCount > 0 Then
            Dim img As Image
            imageCount -= 1
            img = Image.FromFile(dirInfo(imageCount))
            Dim bmp As New Bitmap(img, PictureBox1.Width,
PictureBox1.Height)

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        'set PictureBox1 to previous image
        PictureBox1.Image = bmp
    End If
    If TextBox1.Text <> "" Or RichTextBox1.Text <> "" Then
        'set location to beginning
        loc = 0
        'highlight word in richtextbox1 that is related to the image
        highlightText()
    End If
End If
End Sub

Private Sub PictureBox1_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles PictureBox1.Click
    'open new form containing full sized image
    'when the PictureBox1 is clicked
    If imageCount > -1 Then
        Form2.Location = New Point(0, 0)
        Form2.Width = Screen.PrimaryScreen.WorkingArea.Width
        Form2.Height = Screen.PrimaryScreen.WorkingArea.Height
        Dim img As Image
        img = Image.FromFile(dirInfo(imageCount))
        Form2.BackgroundImage = img
        Form2.Visible = True
    End If

End Sub

Private Function getImageName(ByVal fullName As String)
    'get the name of the image in order to find the word
    'in the richtextbox1
    Dim index, index2 As Integer
    Dim name As String = ""
    For index = 0 To (fullName.Length() - 1)
        If fullName.Chars(index) <> "\" And fullName.Chars(index) <> ":"
Then
            name += fullName.Chars(index)
        Else
            name = ""
        End If
    Next
    Dim finalName As String = ""
    name = name.Remove(name.Length() - 4)

    If name.Contains("_") Then
        Dim temp As String = ""
        For index2 = 0 To (name.Length() - 1)
            If name.Chars(index2) <> "_" Then ' And name.Chars(index2) <>
"0" And name.Chars(index2) <> "1" And name.Chars(index2) <> "2" And
name.Chars(index2) <> "3" And name.Chars(index2) <> "4" And
name.Chars(index2) <> "5" And name.Chars(index2) <> "6" And
name.Chars(index2) <> "7" And name.Chars(index2) <> "8" And
name.Chars(index2) <> "9" Then
                finalName += name.Chars(index2)
            Else
                Dim index3 As Integer
                For index3 = index2 + 1 To (name.Length() - 1)
                    temp += name.Chars(index3)

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        Next
        index2 = index3
    End If
    Next
    imageNum = Val(temp)
Else
    finalName = name
    imageNum = 0
End If

Return finalName
End Function
Private Function highlightText() As String
    'highlight text in richtextbox1
    Dim name As String
    name = getImageName(dirInfo(imageCount))
    Dim nameLength As Integer = name.Length()
    'clear the old highlight so we can highlight the
    'next word
    RichTextBox1.SelectionBackColor = Color.White
    RichTextBox1.SelectionLength() = name.Length
    RichTextBox1.SelectionStart() = clearLocation

    'get the location of the word to highlight
    loc = FindMyText(name, 0)
    While imageNum > 1 And loc <> -1
        'if the next word is the same as the last
        'start searching after the location of the
        'first word. So if the dangerWord was "bad"
        'then the first image was bad_1.jpg. If bad
        'was also the next image (bad_2.jpg) we want
        'the second time bad appears in richtextbox1
        loc = FindMyText(name, loc + nameLength)
        If loc <> -1 Then
            clearLocation = loc
        End If
        imageNum = imageNum - 1
    End While
    'highlight word
    RichTextBox1.SelectionBackColor = Color.DeepSkyBlue
    RichTextBox1.SelectionLength() = name.Length
    RichTextBox1.SelectionStart() = loc
    Return name
End Function
' the following function is from MSDN at http://msdn.microsoft.com/en-us/library/9be0980x.aspx
Private Function FindMyText(ByVal text As String, ByVal start As Integer)
    As Integer
        ' Initialize the return value to false by default.
        Dim returnValue As Integer = -1

        ' Ensure that a search string has been specified and a valid start
point.
        If text.Length > 0 And start >= 0 Then
            ' Obtain the location of the search string in richTextBox1.
            Dim indexToText As Integer = RichTextBox1.Find(text, start, _
                RichTextBoxFinds.MatchCase)

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        ' Determine whether the text was found in richTextBox1.
        If indexToText >= 0 Then
            returnValue = indexToText
        End If
    End If

    Return returnValue
End Function

Private Sub Form1_Resize(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MyBase.Resize
    'keep/adjust positions of items on GUI if the user wants
    'to resize the form
    PictureBox1.Height = RichTextBox1.Height
    PictureBox1.Width = (Me.Width / 2) - 25
    PictureBox1.Location() = New Point(Int(Me.Width / 2),
PictureBox1.Location.Y)

    RichTextBox1.Width = PictureBox1.Width + 20
    RichTextBox1.Anchor = AnchorStyles.Bottom + AnchorStyles.Left +
AnchorStyles.Right + AnchorStyles.Top
    Label2.Anchor = AnchorStyles.Top
    Label2.Location = New Point(PictureBox1.Left + 10, Label2.Location.Y)
    TextBox2.Anchor = AnchorStyles.Top
    TextBox2.Location = New Point(PictureBox1.Left + 10,
TextBox2.Location.Y)
    Button4.Anchor = AnchorStyles.Top
    Button2.Location = New Point(TextBox2.Right + 6, Button2.Location.Y)
    Button1.Anchor = AnchorStyles.Bottom
    Button2.Anchor = AnchorStyles.Bottom
    Button1.Location = New Point((PictureBox1.Right / 2) + 70,
Button1.Location.Y)
    Button2.Location = New Point(PictureBox1.Right - 100,
Button2.Location.Y)
    If PictureBox1.Image Is Nothing Then
    Else
        Dim img As Image
        img = Image.FromFile(dirInfo(imageCount))
        Dim bmp As New Bitmap(img, PictureBox1.Width, PictureBox1.Height)
        PictureBox1.Image = bmp
    End If

End Sub

Private Sub ClearAllEntriesToolStripMenuItem_Click(ByVal sender As
System.Object, ByVal e As System.EventArgs) Handles
ClearAllEntriesToolStripMenuItem.Click
    TextBox1.Text = ""
    TextBox2.Text = ""
    RichTextBox1.Text = ""
    PictureBox1.Image = Nothing
End Sub

Private Sub ExitToolStripMenuItem_Click(ByVal sender As System.Object,
ByVal e As System.EventArgs) Handles ExitToolStripMenuItem.Click
    Me.Close()
    If (HelpForm.Visible) Then

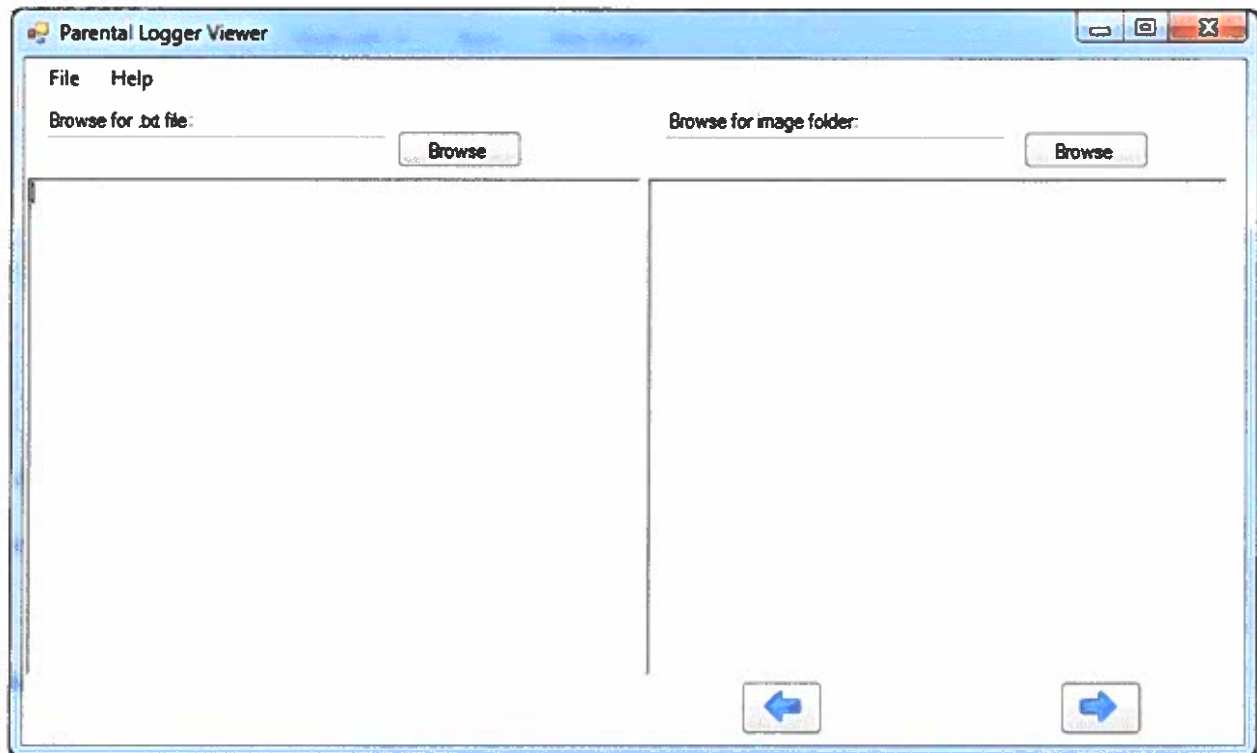
```

```
        HelpForm.Close()
    End If
    If (Form2.Visible) Then
        Form2.Close()
    End If
End Sub

Private Sub AboutToolStripMenuItem_Click(ByVal sender As System.Object,
ByVal e As System.EventArgs) Handles AboutToolStripMenuItem.Click
    MsgBox("Parental Key Logger Viewer" & (Chr(13)) & "2011 Christopher
Wyant", vbOKOnly, "About")
End Sub

Private Sub ViewHelpToolStripMenuItem_Click(ByVal sender As
System.Object, ByVal e As System.EventArgs) Handles
ViewHelpToolStripMenuItem.Click
    HelpForm.Visible = True
End Sub
End Class
```

Project Code: ParentalLoggerViewer- Form1.vb (DESIGN)



Project Code: ParentalLoggerViewer- HelpForm.vb

```

Public Class HelpForm
    Private Sub ListBox2_SelectedIndexChanged(ByVal sender As System.Object,
        ByVal e As System.EventArgs) Handles ListBox2.SelectedIndexChanged
        If (ListBox2.SelectedItem = "Introduction") Then
            ListBox1.Items.Clear()
            ListBox1.Items.Add("Parental Key Logger Viewer:")
            ListBox1.Items.Add(" ")
            ListBox1.Items.Add(" ")
            ListBox1.Items.Add("Welcome to the Parental Key Logger Viewer!
This setup program is used to view")
            ListBox1.Items.Add("the results from the Parental Key Logger. To
use this program you must")
            ListBox1.Items.Add("first receive an e-mail from the Parental Key
Logger. The e-mail should")
            ListBox1.Items.Add("contain a zipped folder containing images and
a .txt file containing the")
            ListBox1.Items.Add("keys that were pressed on the child's
computer.")
            ListBox1.Items.Add(" ")
            ListBox1.Items.Add(" ")
            ListBox1.Items.Add("The Help menu contains information that is
intended to help you use the")
            ListBox1.Items.Add("Parental Key Logger Viewer. To use Help,
select a category from the left.")
            ListBox1.Items.Add(" ")
            ListBox1.Items.Add(" ")
            ListBox1.Items.Add("Thank you for using Parental Key Logger")
            ListBox1.Items.Add(" ")
            ListBox1.Items.Add(" ")
            ListBox1.Items.Add("@Christopher Wyant")

        End If
        If (ListBox2.SelectedItem = "Where do I get the files?") Then
            ListBox1.Items.Clear()
            ListBox1.Items.Add("Where do I get the files?")
            ListBox1.Items.Add(" ")
            ListBox1.Items.Add(" ")
            ListBox1.Items.Add("You should receive an e-mail from the
Parental Key Logger. The e-mail will be sent")
            ListBox1.Items.Add("to the e-mail address you provided in the
Start Logger program. This e-mail contains")
            ListBox1.Items.Add("the necessary files to use the Parental Key
Logger Viewer. One file should be a")
            ListBox1.Items.Add("zipped folder that contains images and the
other file should be a .txt file that")
            ListBox1.Items.Add("contains the keys that were pressed by the
child. You will need to save these files")
            ListBox1.Items.Add("to your computer.")

        End If
        If (ListBox2.SelectedItem = "Browse for .txt file") Then
            ListBox1.Items.Clear()
            ListBox1.Items.Add("Browse for .txt file:")
            ListBox1.Items.Add(" ")
        End If
    End Sub
End Class

```

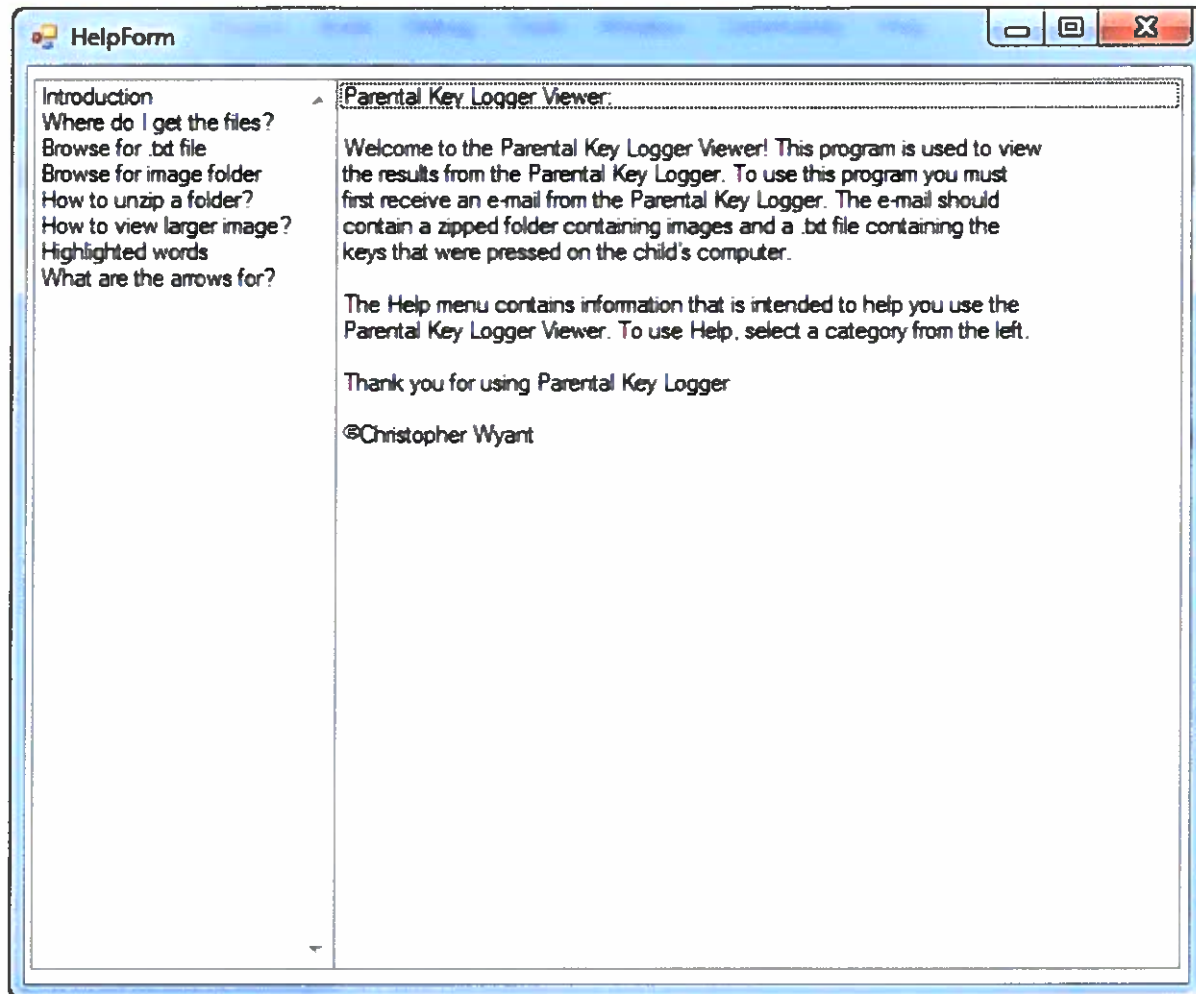


```
ListBox1.Items.Add(" ")
ListBox1.Items.Add("After you have saved the necessary files to
your computer (see Where do")
ListBox1.Items.Add("I get the files?) you can select the Browse
for text file button. Use ")
ListBox1.Items.Add("the directory to find the .txt file you saved
from the e-mail to your")
ListBox1.Items.Add("computer. Once you find the file select it
and click the open button.")
ListBox1.Items.Add("The .txt file should now be loaded into the
program.")
End If
If (ListBox2.SelectedItem = "Browse for image folder") Then
ListBox1.Items.Clear()
ListBox1.Items.Add("Browse for image folder:")
ListBox1.Items.Add(" ")
ListBox1.Items.Add(" ")
ListBox1.Items.Add("After you have saved the necessary files to
your computer (see Where do")
ListBox1.Items.Add("I get the files?) you must unzip the image
folder (see How to unzip a")
ListBox1.Items.Add("folder?). You can now select the Browse for
image folder button. Use ")
ListBox1.Items.Add("the directory to find the unzipped image
folder you saved from the")
ListBox1.Items.Add("e-mail to your computer. Once you find the
folder select it and click")
ListBox1.Items.Add("the open button. The first image in this
folder should now be loaded")
ListBox1.Items.Add("into the program.")
End If
If (ListBox2.SelectedItem = "How to unzip a folder?") Then
ListBox1.Items.Clear()
ListBox1.Items.Add("How to unzip a folder?")
ListBox1.Items.Add(" ")
ListBox1.Items.Add(" ")
ListBox1.Items.Add("After you have saved the necessary files to
your computer (see Where do")
ListBox1.Items.Add("I get the files?) you must unzip the image
folder. To do this you may")
ListBox1.Items.Add("need to download an unzipping program like
7zip. Once you have a program")
ListBox1.Items.Add("to unzip folders, right click the zipped
image folder that you saved from")
ListBox1.Items.Add("the e-mail. Scroll over your unzipping
program (7-Zip) and select 'Extract")
ListBox1.Items.Add("Here'. You should now be ready to browse and
select for the image folder.")
ListBox1.Items.Add(" ")
ListBox1.Items.Add(" ")
ListBox1.Items.Add("Note: make sure when you open the image
folder you can see the images and")
ListBox1.Items.Add("not another folder. If you see a folder
inside this inner folder will be the")
ListBox1.Items.Add("folder you must select when browsing for the
image folder ")
End If
```

```

    If (ListBox2.SelectedItem = "How to view larger image?") Then
        ListBox1.Items.Clear()
        ListBox1.Items.Add("How to view larger image?")
        ListBox1.Items.Add(" ")
        ListBox1.Items.Add(" ")
        ListBox1.Items.Add("There are two ways to view the image at a
larger size. The first way is by")
        ListBox1.Items.Add("maximizing the main window. The second way is
to click the image.")
    End If
    If (ListBox2.SelectedItem = "Highlighted words") Then
        ListBox1.Items.Clear()
        ListBox1.Items.Add("Highlighted words:")
        ListBox1.Items.Add(" ")
        ListBox1.Items.Add(" ")
        ListBox1.Items.Add("The highlighted word in the text box is the
danger word related to the image.")
        ListBox1.Items.Add("The danger word is the word that the child
typed that was inappropriate and")
        ListBox1.Items.Add("caused the image of the child's screen to be
taken.")
    End If
    If (ListBox2.SelectedItem = "What are the arrows for?") Then
        ListBox1.Items.Clear()
        ListBox1.Items.Add("What are the arrows for?")
        ListBox1.Items.Add(" ")
        ListBox1.Items.Add(" ")
        ListBox1.Items.Add("The arrows at the bottom of the window are
used to view the other images if there are")
        ListBox1.Items.Add("multiple images to view. The right arrow will
display the next image and highlight the")
        ListBox1.Items.Add("danger word related to this image. If there
are no more images then nothing will change.")
        ListBox1.Items.Add("The left arrow will display the previous
image and highlight the danger word related to")
        ListBox1.Items.Add("this image. If you are on the first image
then nothing will change. ")
    End If
End Sub
End Class

```

Project Code: ParentalLoggerViewer- HelpForm.vb (DESIGN)

Project Code: uninstall- Form1.vb

```
Public Class Form1

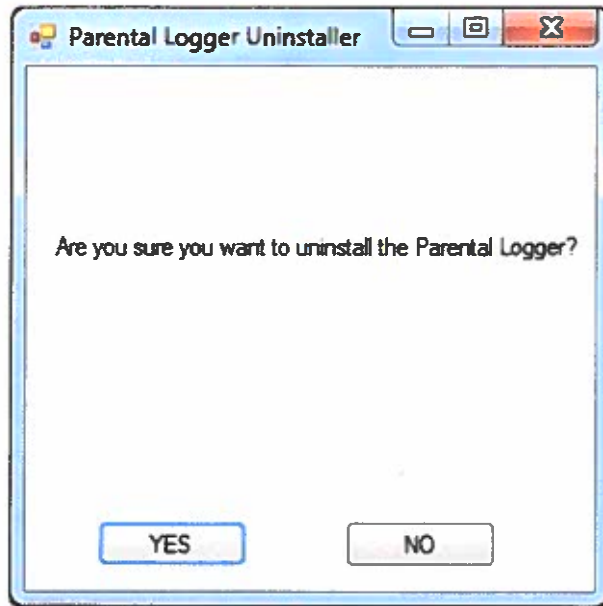
    Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MyBase.Load

        End Sub

        Private Sub Button2_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles Button2.Click
            Me.Close()
        End Sub

        Private Sub Button1_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles Button1.Click
            Me.Visible() = False
            Form2.Visible() = True
            Form2.ProgressBar1.Increment(25)
            'to delete the registry key
            Dim registryKey As Microsoft.Win32.RegistryKey
            registryKey =
Microsoft.Win32.Registry.CurrentUser.OpenSubKey("SOFTWARE\Microsoft\Windows\C
urrentVersion\Run", True)
            Form2.ProgressBar1.Increment(25)
            registryKey.DeleteValue("Logger", False)
            Form2.ProgressBar1.Increment(25)
            registryKey.Close()
            Form2.ProgressBar1.Increment(25)
            Form2.Label1.Text() = "Uninstall Completed"

        End Sub
End Class
```

Project Code: uninstall- Form1.vb (DESIGN)

Project Code: uninstall- Form2.vb

```
Public Class Form2

    Private Sub Form2_FormClosed(ByVal sender As System.Object, ByVal e As
System.Windows.Forms.FormClosedEventArgs) Handles MyBase.FormClosed
        Form1.Close()
    End Sub
End Class
```

Project Code: uninstall- Form2.vb (DESIGN)

APPENDIX B: Senior Defense Presentation

Senior Defense Presentation

Parental Key Logger

Christopher Wyant

Advisor: Dr. Valerie Sessions

Introduction

- Parents have difficulty keeping track of things their child does on a computer
- Create a Key Logger specific for parents to use with their children
- Why a Key Logger?
 - Beneficial to parents
 - Something I could use when I have children

Senior Defense Presentation

Background

- Parents are able to track what their children are doing
- Help parents be aware of what personal info their children are releasing online
- Give proof against false accusations
- Create a program that is easy to use

BACKGROUND

- Track:
 - Keys pressed
 - Take screen shots
 - When "Danger Words" are typed
 - When an address/personal info is entered into a form
 - Email results to parent

Senior Defense Presentation

Standard Danger Words List

shut	stupid	dumb	frickin	frick
fudge	jeeze	jeeze	gosh	darn
shoot	crap	dang	hate	sucks
suck	jerk	butt	retard	tard
gay	lesbian	homosexual	homo	idiot
dookie	poop	heck	hell	blame
blamed	sneak	steal	stole	rob
robbed	cheat	cheater	cheated	smoke
drugs	meet	boy	girl	man
woman	male	female	leave	run
away	love	online	games	alone
suicide	trashed	drunk	pick	up
homework				

Contact Information cancel

Email: lock

lock

Add / Manage Emails

IM Screen Name(s): add lock

Add another screen name

Mobile Phone: lock

Country: + and country codes

Land Phone: lock

Address: lock

City/Town:

Neighborhood:

Zip:

School Mailbox:

Residence:

Room:

Website: lock

Create a Facebook badge for your website

Senior Defense Presentation

Theory

- **Parental Setup GUI**
 - Allow parents to decide the Danger Words for customization
- **Logging**
 - Track keys typed
 - Screen capture based off specific words typed
 - Email .log and screen captures to parent

Theory

- **Parental Logger Viewer**
 - Easy to use interface to view results (screen captures and .log file)

Senior Defense Presentation

Regular Expressions

```

1518 Regex phoneNum="(\\d{3}|\\d{4})\\d{3}|\\d{10}"//series of numbers in form 1000100000000000
1519 //represents phone number with area code and w/o hyphen or 1000000000000000 w/o area code and w/o hyphen
1520
1521 Regex hyperPhone="(\\d{3}|\\d{4})\\d{3}|\\d{10}|(\\d{3}|\\d{4})\\d{3}"//series of numbers in form
1522 1000100000000000 or 1000100000000000 or 1000100000000000 or 1000100000000000 which represents phone number
1523 //both area code and hyphen in 1000100000000000 area code and both hyphen
1524
1525 Regex sen="(\\d{3}|\\d{4})\\d{3}"//series of numbers in form 1000100000000000 which represents sen w/o hyphen or
1526 //10000 less four digits of sen
1527
1528 Regex hyperSen="(\\d{3}|\\d{4})\\d{3}|\\d{10}"//series of numbers in form 1000100000000000
1529 //represents full sen with hyphens

```

Demo

Senior Defense Presentation

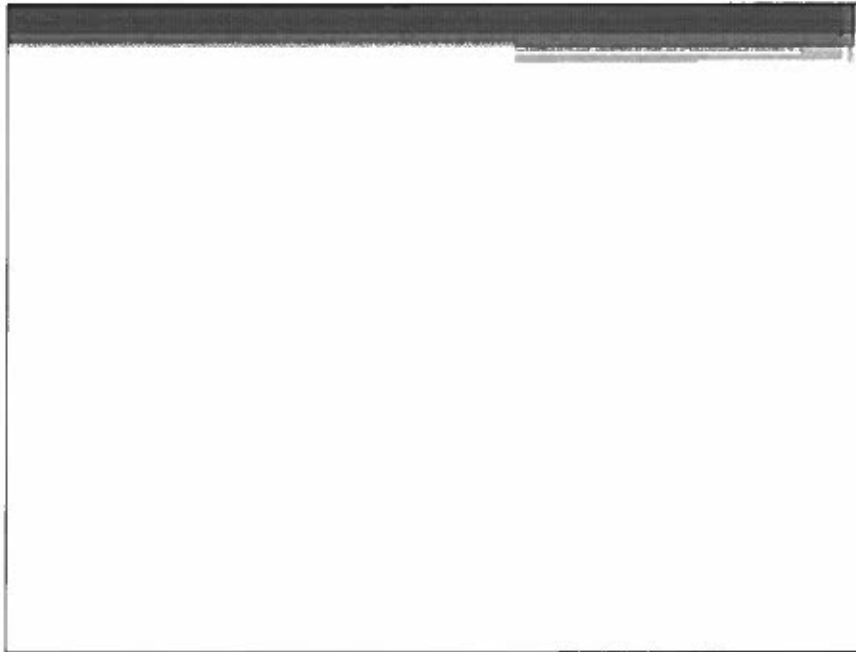
Summary

- Parents need an easy way to keep track of children's actions on the computer
- Allow parents to add their own Danger Words to allow customization and expand the standard list
- Warn parents about personal info entered into the computer
 - Phone numbers
 - Address

Future Work

- Expand the standard set of the Danger Words
- Test on multiple Operating Systems to determine official system requirements
- Use multithreading

Senior Defense Presentation



Setup GUI

A screenshot of a Windows-style application window titled "Parental Logger Setup". The window has a menu bar with "File" and "Help". It contains several input fields: "Your Email address", "Child's Name", "Address", "City", and "State" (a dropdown menu). Below these is a "Email Frequency" section with radio buttons for "1", "5", and "10". At the bottom is a "Danger Words" text area and a "Start" button.

Parental Logger Setup

File Help

Your Email address

Child's Name

Address

City State

Email Frequency 1 5 10

Danger Words

Start

Senior Defense Presentation

Setup GUI: Help Menu

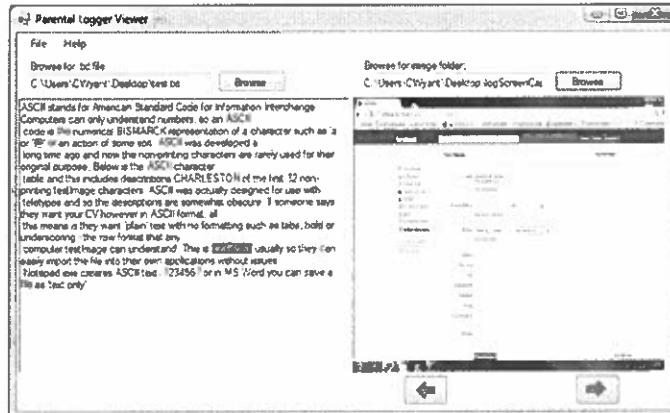


Parental Logger Viewer



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Parental Logger Viewer

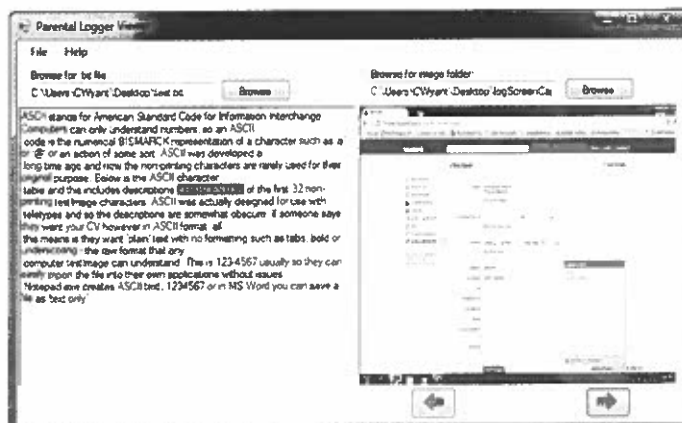


Parental Logger Viewer

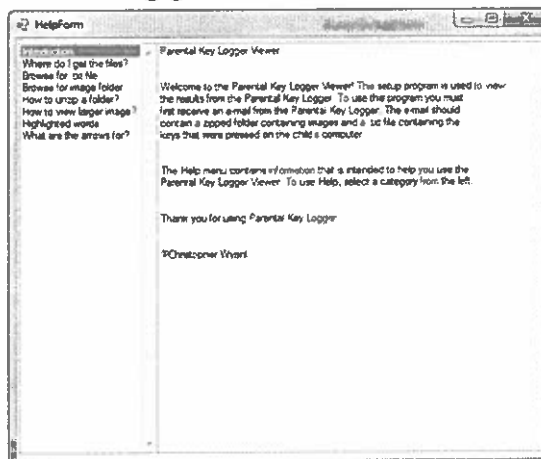


Senior Defense Presentation

Parental Logger Viewer



Parental Logger Viewer Help



Senior Defense Presentation

Problems Encountered and Solutions

- **Determine Logging Method**
 - GetAsyncKeyState()
 - Low-Level Keyboard Hook
- **Danger Words List**
 - Creating an effective list which is also CSU appropriate
 - Solution: allow parents to add Danger Words to the standard list at Setup

Problems Encountered and Solutions

- **Startup**
 - Program must start automatically with windows startup programs
 - Solution: Create a Registry Key
- **Screen Capture**
 - In C++ simulate the press of "PrtSc" Button but could not save image from system clipboard
 - Solution: Use Visual Basic's Graphics CopyFromScreen() Method

Senior Defense Presentation

Problems Encountered and Solutions

- **Email**
 - Find a SMTP (Simple Mail Transfer Protocol) server
 - Solution: Create a Gmail account
- **Run in Background**
 - C++ launches window before Main() is called
 - Solution: Use WINAPI WinMain()

Problems Encountered and Solutions

- **Testing**
 - Finding and fixing issues when performing tests with the executables
 - Could not use Visual Studio to debug
 - Tedious
 - Edit code->recompile->restart computer->Test and see if change worked.